Information Technology University



Lab Manual

for

CS220: Artificial Intelligence

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Semester Fall 2023 : **3**
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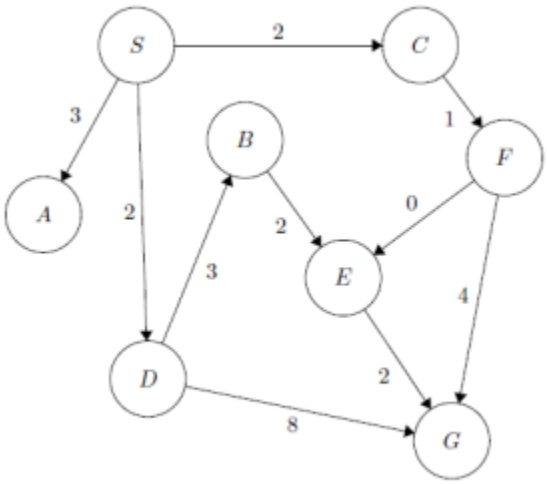
Lab - 06

Exercise 1:

a. **Breadth-First Search (BFS)**
b. **Depth-First Search (DFS)**

For both of the above algorithms:

1. Consider the graph given below, with the start state `'S'` and goal state `'G'`.
2. Implement the algorithms to find the **path** from `'S'` to `'G'`.
3. Return the **path taken** and the **cost** associated with that path for each algorithm.



Exercise 2:

Uniform Cost Search (UCS)

Implement the **Uniform Cost Search (UCS)** algorithm to find the least-cost path from node `'S'` to node `'G'` in the weighted graph given in **Exercise 1**. Your implementation should return:

1. The **path taken** from node `'S'` to node `'G'`.
2. The **cost** associated with that path.

Hint: You can use the **Priority Queue** data structure to efficiently retrieve the node with the least cost at each step. For this, you can import the `PriorityQueue` class as shown below:

```
from queue import PriorityQueue
```